

Recall .0.1

Let L/K be field extension of number fields. Say L/K is Galois if L is the splitting field of some polynomial in $f \in K[x]$, and $[L : K] = \deg(f)$.

In this field extension of number field K , we have a similar factorization. For any prime ideal $\mathfrak{p}$ of $\mathcal{O}_K$, we have

$$\mathfrak{p}\mathcal{O}_L = Q_1^{e_1} Q_2^{e_2} \cdots Q_r^{e_r},$$

where Q_i are distinct prime ideals of $\mathcal{O}_L$, and e_i are positive integers. A similar relation between n, e, f, r holds:

$$n = \sum_{i=1}^r e_i f_i$$

where $f_i = [\mathcal{O}_L/Q_i : \mathcal{O}_K/\mathfrak{p}]$.

A similar proof can be applied to get the following results:

- In our setting, L/K is Galois, since our previous proof did not rely on the base field, we still have $e_1 = e_2 = \cdots = e_r$ and $f_1 = f_2 = \cdots = f_r$.
- We have some similar definitions:
 - $G := \text{Gal}(L/K) = \text{Aut}_K(L)$
 - $D := D(Q_i/\mathfrak{p}) = \{\sigma \in G \mid \sigma(Q_i) = Q_i\}$
 - This decomposition group yields the following surjective homomorphism:

$$D(Q_i/\mathfrak{p}) \twoheadrightarrow \text{Aut}_{\mathcal{O}_K/\mathfrak{p}}(\mathcal{O}_L/Q_i)$$

- Let $I := I(Q_i/\mathfrak{p})$ be the kernel of this homomorphism, then

$$D/I \cong \text{Aut}_{\mathcal{O}_K/\mathfrak{p}}(\mathcal{O}_L/Q_i) \cong \mathbb{Z}/f_i\mathbb{Z}$$

- $|I| = e, |D| = ef, |G| = efr$.

Theorem .0.1

Let L/K be a finite Galois field extension, let

$$G = \text{Gal}(L/K)$$

Then there is an order-reversing bijection

$$\left\{ \begin{array}{l} \text{fields } M \text{ with} \\ K \subseteq M \subseteq L \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} \text{subgroups of } G \end{array} \right\}$$

$$\{\sigma \in G : \sigma|_M = \text{id}_M\} \longrightarrow M,$$

$$L^H = \{x \in L : \sigma(x) = x \text{ for all } \sigma \in H\} \longleftarrow H.$$

and $[L : M]$ is the size of the corresponding group for M .

Definition .0.1

Let D be the decomposition group of $Q_i/\mathfrak{p}$. I the corresponding inertia group. Call L^D the decomposition field of Q_i , and L^I the inertia field of Q_i .

The following diagram characterizes the relation between subgroups in G and subfields in L .

$$\begin{array}{ccc}
 L = L^1 & \leftrightarrow & 1 \\
 | & & | \\
 e & & e \\
 | & & | \\
 L^I & \leftrightarrow & I \\
 | & & | \\
 f & & f \\
 | & & | \\
 L^D & \leftrightarrow & D \\
 | & & | \\
 r & & r \\
 | & & | \\
 K = L^G & \leftrightarrow & G
 \end{array}$$

Lemma .0.2

For any subgroup H of G , let $K' = L^H$ and $\mathfrak{p}' = \mathfrak{p} \cap \mathcal{O}_{K'}$, then

$$D(Q_i/\mathfrak{p}') = D \cap H$$

$$I(Q_i/\mathfrak{p}') = I \cap H$$

Proposition .0.3

L^I is the largest field K' such that $K \subset K' \subset L$ and $e(\mathfrak{p}/\mathfrak{p}') = 1$. L^D is the largest field K' such that $K \subset K' \subset L$ and $e(\mathfrak{p}/\mathfrak{p}') = 1$ and $f(\mathfrak{p}/\mathfrak{p}') = 1$. i.e. For $K \subset K' \subset L$, and $e = f = 1$, then $K' \subset L^D$.

Proof. Let $K' = L^I$, and let $I' = I(Q_i/\mathfrak{p}')$. Then the lemma tells us that $I' = I \cap I = I$. By HW problem,

$$e(Q_i/\mathfrak{p}) = e(Q_i/\mathfrak{p}') \cdot e(\mathfrak{p}/\mathfrak{p}')$$

Since $e(Q_i/\mathfrak{p}) = |I|$, and $e(Q_i/\mathfrak{p}') = |I'| = |I|$, we get $e(\mathfrak{p}/\mathfrak{p}') = 1$.

For any K' such that $K \subset K' \subset L$ such that $e(\mathfrak{p}'/\mathfrak{p}) = 1$, $K' = L^H$ for some subgroup H of G . Since $I' = I \cap H$,

$$L^{I'} = L^{I \cap H} = L^I \cdot L^H = L^I \cdot K'$$

Using the assumption that $e(\mathfrak{p}'/\mathfrak{p}) = 1$,

$$\begin{aligned}
 e(Q_i/\mathfrak{p}) &= e(Q_i/\mathfrak{p}') \cdot e(\mathfrak{p}'/\mathfrak{p}) \\
 &= e(Q_i/\mathfrak{p}')
 \end{aligned}$$

So $|I| = |I'|$. Therefore, $I' = I$.

Proof continued next time.

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